



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

UNIT TEST

17-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TNAS]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	B	C	C				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B,C	A,C,D	A,D	B,C	C,D	A,C,D	A,C,D	A,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	24.00	9.00	15.00	7.00	8.00	0.00		

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4				
	A.	A	C	D	A				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,B	A,C,D	B,C	A,B,C,D	A,B,C,D	B,D	A,C,D	A,C
SECTION-II	Q.	1	2	3	4	5	6		
	A.	3.20	5.00	0.58 or 0.59	0.16 or 0.17	5.00	4.00		

PART-3 : MATHEMATICS

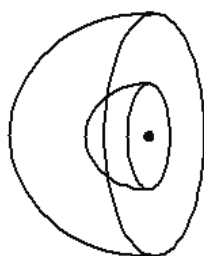
SECTION-I (i)	Q.	1	2	3	4				
	A.	A	B	C	B				
SECTION-I (ii)	Q.	5	6	7	8	9	10	11	12
	A.	A,C,D	A,B	C,D	B	A,C	B,C,D	A,C	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6		
	A.	1.00	6.00	0.25	1.00	1.87 or 1.88	3.00		

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A)



To calculate initial potential energy of the particle, first a thin hemispherical shell of radius x and radial thickness dx is considered.

Volume of material of the shell = $2\pi x^2 dx$

charge on shell is $dQ = \rho(2\pi x^2 dx)$

Potential energy of the particle, due to charge of the shell considered is

$$dU = \frac{-qdQ}{4\pi\epsilon_0 x} = \frac{-q\rho x dx}{4\epsilon_0}$$

$$\Rightarrow U = \int_{x=0}^{x=R} \frac{-q\rho x dx}{2\epsilon_0} = \frac{-q\rho R^2}{4\epsilon_0}$$

Finally potential energy is zero.

$$\text{Thus, workdone} = \Delta U = \frac{q\rho R^2}{4\epsilon_0}$$

2. Ans (B)

$$E = -y\hat{i} - x\hat{j}$$

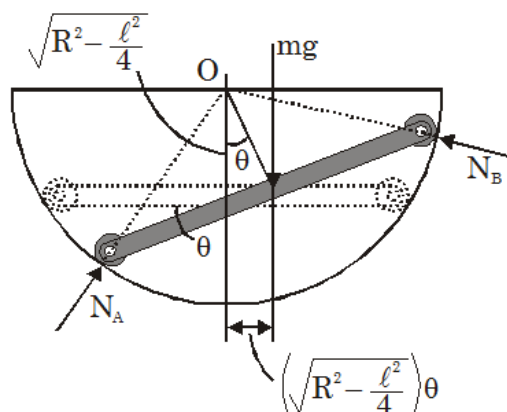
$$\tan \theta = \frac{E_y}{E_x} = \frac{x}{y}$$

3. Ans (C)

$$\text{Charge on inner most shell is } - \left(\frac{r_{\text{inner}}}{r_{\text{outer}}} \right) Q.$$

4. Ans (C)

Moment of inertia about point O :



$$I_0 = \frac{1}{12} m l^2 + m \left(\sqrt{R^2 - \frac{l^2}{4}} \right)^2$$

$$= m \left(R^2 - \frac{1}{6} l^2 \right)$$

$$O + \Sigma M_0 = I_0 \alpha; \quad mg \left(R^2 - \frac{l^2}{4} \right) \theta = -m \left(R^2 - \frac{1}{6} l^2 \right) \theta$$

$$\ddot{\theta} + \frac{3g(4R^2 - l^2)^{1/2}}{6R^2 - l^2} \theta = 0$$

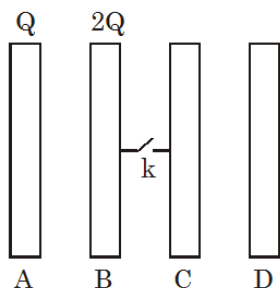
From the above differential equation,

$$\omega_n = \sqrt{\frac{3g(4R^2 - l^2)^{1/2}}{6R^2 - l^2}}$$

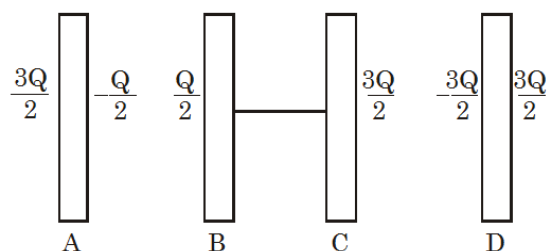
PART-1 : PHYSICS

SECTION-I (ii)

6. Ans (A,C,D)



After switch on final charge distribution is like this



final charge on C is $\frac{3Q}{2}$

final charge on B is $\frac{Q}{2}$

7. Ans (A,D)

$$\frac{dy}{dx} = -\frac{40}{6400}x; \quad \frac{d^2y}{dx^2} = -\frac{40}{6400}$$

$$\tan \theta = \frac{dy}{dx} = -\frac{40}{6400} \times 80 \quad (\text{At A, } x = 80)$$

$$= -0.5$$

$$\therefore |\cos \theta| = \frac{2}{\sqrt{5}}, \quad |\sin \theta| = \frac{1}{\sqrt{5}}$$

Radius of curvature

$$= \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}{\left| \frac{d^2y}{dx^2} \right|} = \frac{[1 + (0.5)^2]^{3/2}}{\frac{40}{6400}} = 223.6$$

$$\therefore N = mg \cos \theta - \frac{mv^2}{R} = 6.865 \text{ kN}$$

$$f = mg \sin \theta - ma_t = 1.18 \text{ kN}$$

8. Ans (B,C)

$$y = 0.2 e^x$$

$$\text{At } y = 0.2, \quad x = 0.$$

$$\tan \theta = 0.2 \text{ at } x = 0$$

$$\sin \theta = \frac{1}{\sqrt{26}}, \quad \cos \theta = \frac{5}{\sqrt{26}}$$

$$\therefore \frac{dy}{dx} \Big|_{x=0} = 0.2; \quad \frac{d^2y}{dx^2} \Big|_{x=0} = 0.2$$

$\therefore$ Radius of curvature

$$= \frac{\left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{3/2}}{\frac{d^2y}{dx^2}} = \frac{(1 + 0.2^2)^{3/2}}{0.2} = 5.3$$

$$\therefore N - mg \cos \theta$$

$$= \frac{mv^2}{R} \Rightarrow N = mg \cos \theta + \frac{mv^2}{R}$$

$$\therefore N = 80 \times \frac{5}{\sqrt{26}} + \frac{8 \times 1.5^2}{5.3} = 81.84$$

$$\text{Also tangential acceleration} = g \sin \theta = 1.96 \text{ m/s}^2$$

9. Ans (C,D)

Let us first notice that the image of the point light source appears at the following distance from the lens:

$$\ell = \left(\frac{1}{36} - \frac{1}{60} \right)^{-1} \text{ mm} = 90 \text{ mm},$$

meaning on the screen. If there were two plane mirrors in place of the reflecting cylindrical walls then an infinite sequence of the light source's images would appear (reflection, reflection's reflection and so on), where the distance between two neighboring images is equal to the distance between the mirrors, 12 mm. Let us now observe the cylindrical case. A ray lying on the plane of the figure has the same path as it would have in the case of a flat mirror, meaning that on the plane of the figure the images also occur in a periodical sequence where the distance between the images is 12 mm. The plane of the figure can be randomly turned around the symmetry axis of the system; this means that the images are actually "smudged" along a concentric circle of radius $n \times 12 \text{ mm}$ where n is an integer. The lens creates a $\frac{90}{60}$ times magnified image of these circles on the screen where the radius of the concentric circles is $R = n \times 18 \text{ mm}$.

10. Ans (A,C,D)

Stretch in spring = y

$$y = \ell (1 - \cos\theta)$$

$$U = \frac{1}{2}ky^2 + mg \left[\frac{\ell \cos\theta}{2} + y - \frac{\ell}{2} \right]$$

$$\frac{dU}{d\theta} = 0 \Rightarrow \theta = 60^\circ \& \frac{d^2U}{d\theta^2} > 0$$

11. Ans (A,C,D)

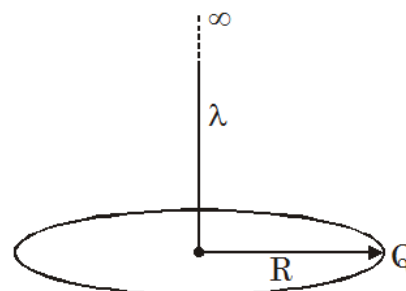
Electric field due to a point charge is non-uniform.

$\therefore$ Force exerted by dipole on point charge

$$= \frac{QkP}{r^3} \sqrt{1 + 3\cos^2\theta}$$

Hence system can't be in equilibrium.

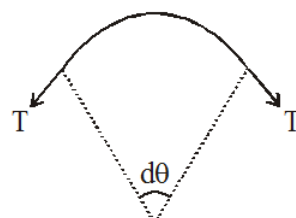
12. Ans (A,D)



Electric field due to wire at ring will have two component is horizontal $\left(\frac{k\lambda}{R} \right)$ & in vertical $\left(\frac{k\lambda}{R} \right)$.

$$\therefore \text{Net force experience by ring} = \frac{k\lambda}{R} Q = \frac{\lambda Q}{4\pi\epsilon_0 R}$$

Also,



$$2T \frac{d\theta}{2} = \frac{1}{4\pi\epsilon_0} \lambda \frac{Qd\theta}{2\pi R} \Rightarrow T = \frac{\lambda Q}{8\pi^2 \epsilon_0 R}$$

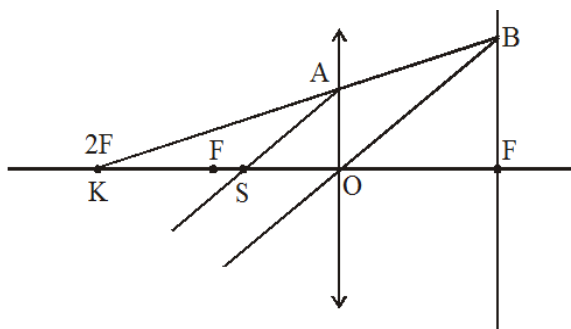
PART-1 : PHYSICS

SECTION-II

1. Ans (24.00)

First solution

If the light source located on the optical axis is at the distance of 18 cm from the lens which is equal to double the focal length then the image of the light source is located on the other side of the lens also at double the focal length from the lens, meaning at the distance 18 cm. For the images of the two light sources to coincide the other image has to be virtual. Let us construct the location of the light source if the location of the image is known.



From the similar triangles ΔKAO and ΔKBF we get that

$$\frac{AO}{BF} = \frac{KO}{KF} = \frac{2f}{3f} = \frac{2}{3}$$

From the other pair of similar triangles ΔSAO and ΔOBF we get that

$$\frac{AO}{BF} = \frac{OS}{FO} = \frac{b}{f}$$

where b is the distance of the second light source from the lens. Connecting the two relations we get

$$\frac{KO}{KF} = \frac{OS}{FO} \Rightarrow \frac{2}{3} = \frac{b}{f} \Rightarrow b = \frac{2f}{3} = 6\text{cm}$$

The second light source has to be located from the lens at a distance 6cm and the distance between the two light sources is $6\text{ cm} + 18\text{cm} = 24\text{ cm}$.

Second solution

Instead of similar triangles we can use the thin lens formula. Knowing that the light source is at the distance $a = 18\text{ cm}$ from the lens and the focal length of the lens is $f = 9\text{ cm}$ we can find the distance k of the image from the lens:

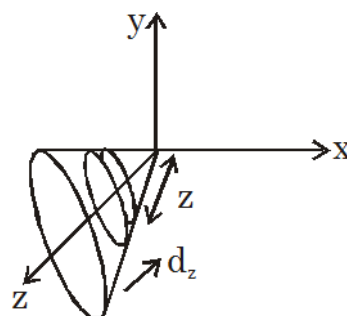
$$\frac{1}{a} + \frac{1}{k} = \frac{1}{f} \Rightarrow k = 18\text{ cm}$$

Since the images of the two light sources have to coincide then the image made by the second light source has to be virtual and the light source has to be located at the same side of the lens as the image. Using the thin lens formula we find the location of the second light source

$$\frac{1}{a} - \frac{1}{k} = \frac{1}{f} \Rightarrow a = 6\text{ cm}$$

The second light source has to be at the distance 6 cm from the lens and the distance between the two light sources is $6\text{ cm} + 18\text{cm} = 24\text{ cm}$.

2. Ans (9.00)



$$d\sigma = \rho dz$$

$$dE = \frac{d\sigma}{2\epsilon_0} (1 - \cos \phi)$$

$$E = \frac{\rho}{2\epsilon_0} \times h (1 - \cos \phi)$$

$$F = \frac{q\rho}{2\epsilon_0} h \left(1 - \frac{1}{\sqrt{2}} \right)$$

$$F = 1 \times \frac{\sqrt{2}}{2\pi\epsilon_0} \times 10^{-6} \times (\sqrt{2} + 1) \frac{(\sqrt{2} - 1)}{\sqrt{2}} q$$

$$= 9 \times 10^9 \times 10^{-6} \times 2 \times 5 \times 10^{-4}$$

$$= 9\text{ N}$$

3. Ans (15.00)

From the given condition,

$$KE = W_{\text{spring}}$$

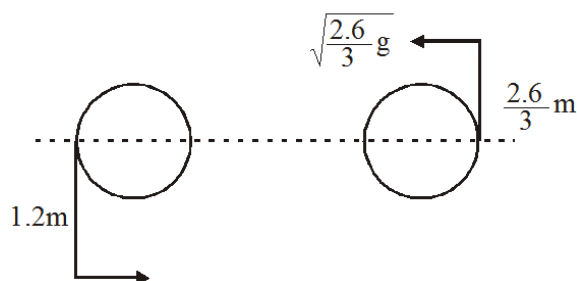
$$\int_0^{0.2} ks^2 ds = \frac{1}{2} Mv^2 - 0$$

$$\Rightarrow \left| \frac{ks^3}{3} \right|_0^{0.2} = \frac{1}{2} Mv^2$$

$$\Rightarrow \frac{k}{3} = 8 \times 10^{-3} = \frac{1}{2} \times 5 \times 10^3 \times 16$$

$$\Rightarrow k = 15 \times 10^6 \text{ N/m}^2$$

4. Ans (7.00)



If string does not get slack in first round, it will never get slack.

$$\left(\sqrt{\frac{2.6}{3}g} \right)^2 = V^2 - 2g \left(1.2 + \frac{2.6}{3} \right)$$

$$\frac{2.6}{3}g + 2g \frac{6.2}{3} = V^2$$

$$V^2 = 5g = 49$$

$$V = 7 \text{ m/s}$$

5. Ans (8.00)

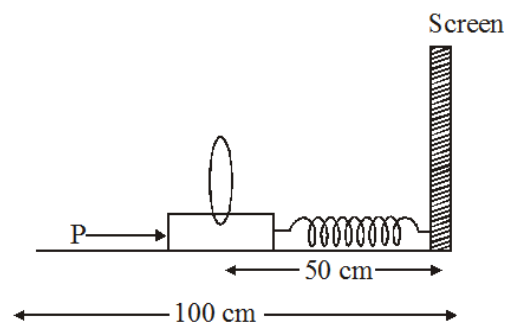
Let the distance of the lens from the object be ℓ when a real image is formed on the screen.

Then

$$\frac{1}{100 - \ell} - \frac{1}{-\ell} = \frac{1}{23}$$

On solving, we get $\ell = (50 \pm 10\sqrt{2}) \text{ cm}$.

Now, if the lens performs SHM and a real image is formed after a fixed time gap, then this time gap must be one-fourth of the time period.



$\therefore$ Phase difference between the two positions of real image must be $\frac{\pi}{2}$. As the two positions are symmetrically located about the origin, phase difference of any of these positions from origin must be $\frac{\pi}{4}$.

$$\Rightarrow 10\sqrt{2}\text{cm} = A \sin \frac{\pi}{4} \Rightarrow A = 20 \text{ cm}$$

To achieve this velocity at the mean position,

$$u_0 = A\omega = A\sqrt{\frac{K}{m}}$$

$$\therefore \text{Required impulse } p = mu_0 = A \sqrt{K/m} = 8 \text{ kg m/s.}$$

6. Ans (0.00)

$$\phi_{\text{net}} = 0$$

$$\therefore Q_{\text{enclosed}} = 0$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. **Ans (A)**

If FeO is like NaCl (rock salt)

Then no. of Fe^{2+} per unit cell will be = 4

crystal $\text{Fe}_{0.75}\text{O}$ contain some Fe^{2+} (x) & some

Fe^{3+} ($0.75 - x$) so $2x + (0.75 - x) \times 3 = 2$

$\Rightarrow x = \frac{1}{4} \Rightarrow$ number of Fe^{2+} per unit cell = 1

4. **Ans (A)**

Fact

PART-2 : CHEMISTRY

SECTION-I (ii)

8. **Ans (A,B,C,D)**

$$3p_y$$



$$(\ell=1) \quad (n=3)$$

(A) Total nodes = $(n - 1) = (3 - 1) = 2 \rightarrow$ Correct

(B) No. of maxima = $(n - 1) = (3 - 1) = 2 \rightarrow$ Correct

(C) n, ℓ and $m \Rightarrow 3, 1, -1$ (Correct)

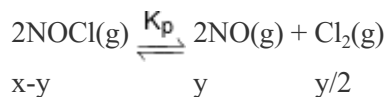
(D) $m \Rightarrow -1$ to $+1$ (any integer value) (Correct)

PART-2 : CHEMISTRY

SECTION-II

1. **Ans (3.20)**

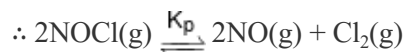
Let $P_{\text{NOCl}} = x \text{ atm}$ $P_{\text{N}_2} = (1 - x) \text{ atm}$



$$x-y \qquad \qquad y \qquad \qquad y/2$$

$$x-y + y + y/2 + (1-x) = 1.2$$

$$\text{or } y = 0.4$$

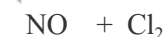


$$\text{At equilibrium } x-0.4 \qquad \qquad 0.4 \qquad \qquad 0.2$$

Now pressure due to added Cl_2 only if no further reaction occurs

$$= 8.3 - 1.2 = 7.1$$

$\therefore$ Pressure due to total Cl_2 if no further reaction occurs = $7.1 + 0.2 = 7.3$



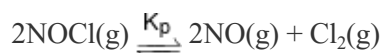
$$x-0.4 \qquad 0.4 \qquad 7.3 \quad p_{\text{total}_1} = 7.3 + x$$

$$x-0.4 + 2z \quad 0.4-2z \quad 7.3-z \quad p_{\text{total}_2} = 7.3 + x-z$$

$$p_{\text{total}_1} - p_{\text{total}_2} = 8.3 - 8.2$$

$$\text{or } 7.3 + x - (7.3 + x - z) = 0.1$$

$$\text{or } z = 0.1$$



$$\text{At equilibrium } x-0.2 \qquad \qquad 0.2 \qquad \qquad 7.2$$

Now $(K_p)_{\text{at equilibrium 1}} = (K_p)_{\text{at equilibrium 2}}$

$$\text{or } \frac{(0.4)^2 \times 0.2}{(x-0.4)^2} = \frac{(0.2)^2 \times (7.2)}{(x-0.2)^2}$$

$$\text{or } \frac{1}{(x-0.4)^2} = \frac{9}{(x-0.2)^2}$$

$$\text{or } \frac{1}{x-0.4} = \frac{3}{(x-0.2)}$$

$$\text{or } 3x - 1.2 = x - 0.2$$

$$\text{or } x = 0.5$$

$$\therefore K_p = \frac{(0.4)^2 \times 0.2}{(0.5-0.4)^2} = \frac{0.032}{0.01} = 3.2$$

2. Ans (5.00)

$$P = KX$$

$$1 = K \times \frac{1 \times 40}{300R \times 1000} \times \frac{100}{18}$$

$$\Rightarrow 1 = K \times \frac{40}{300R} \times \frac{18}{100} \times \frac{1}{1000}$$

$$\Rightarrow K = \frac{300R \times 100 \times 1000}{40 \times 18}$$

$$P = KX$$

$$\text{or, } 4 = \frac{100 \times 300R}{18 \times 40} \times \frac{x}{48 \times \frac{1600}{18}}$$

$$\text{or, } x = \frac{4 \times 40 \times 48 \times 16}{300R \times 1000} = 5g$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A)

$$f(-x) = \frac{-x \cos x}{\left[\frac{-x}{5\pi}\right] + \frac{1}{2}}$$

$$= \frac{-x \cos x}{-\left[\frac{x}{5\pi}\right] - 1 + \frac{1}{2}} = \frac{x \cos x}{\left[\frac{x}{5\pi}\right] + \frac{1}{2}}$$

$$\therefore f(-x) = f(x) \Rightarrow f(x) \text{ is even}$$

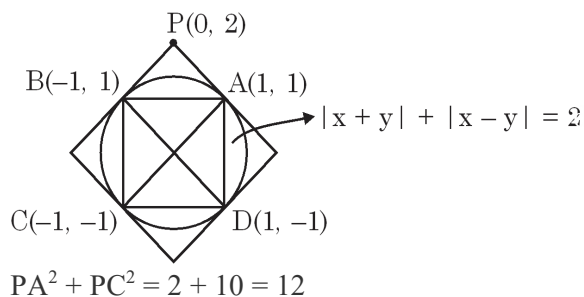
2. Ans (B)

Take L.C.M.

$$\Rightarrow L = \lim_{x \rightarrow 1} \frac{2018 - 2018x^{2019} - 2019 + 2019x^{2018}}{\frac{(1-x^{2018})}{(1-x)} \cdot \frac{(1-x^{2019})}{(1-x)} (1-x)^2}$$

$$\text{apply L.H. rule twice} \Rightarrow L = -\frac{1}{2}$$

3. Ans (C)



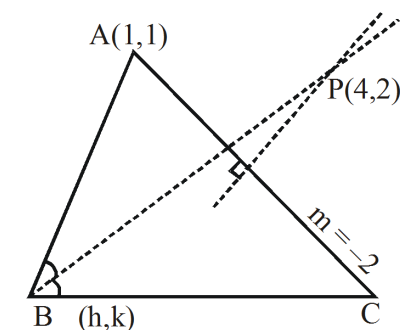
4. Ans (B)

Equation of perpendicular bisector of AC is

$$y - 2 = \frac{1}{2}(x - 4) \text{ i.e. } x - 2y = 0$$

Let $C(\alpha, \beta)$

$\therefore C$ is image of $A(1,1)$ in $x - 2y = 0$



$$\Rightarrow \frac{\alpha - 1}{1} = \frac{\beta - 1}{-2} = -2 \left(\frac{1 - 2}{1 + 4} \right)$$

$$\Rightarrow \alpha = \frac{7}{5}, \beta = \frac{1}{5} \Rightarrow C \left(\frac{7}{5}, \frac{1}{5} \right)$$

again image of A in angle bisector $x - y - 2 = 0$

lies on side BC

$$\Rightarrow \frac{h - 1}{1} = \frac{k - 1}{-1} = \frac{-2(1 - 1 - 2)}{2}$$

$$\Rightarrow h = 3, k = -1 \Rightarrow (3, -1)$$

$$\Rightarrow \text{slope of side BC is } -\frac{3}{4}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

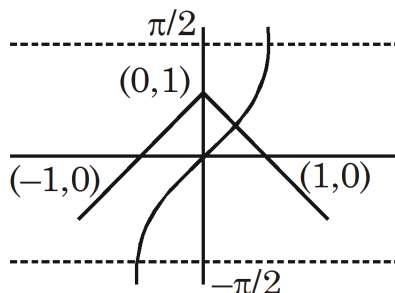
5. Ans (A,C,D)

No. of solutions of equations

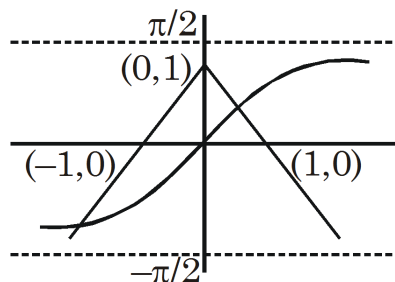
$$\sin^{-1}x + |x| = 1, \tan^{-1}x + |x| = 1$$

and $\sec^{-1}x + |x| = 1$ are x_1, x_2 and x_3 resp. so

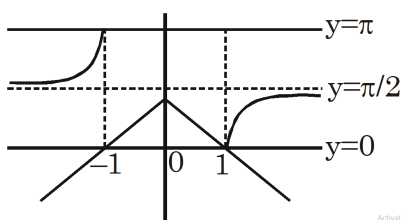
$$n_1 = 1 \quad \text{Plot } \sin^{-1}x \text{ \& } 1 - |x|$$



$$n_2 = 2 \quad \text{Plot } \tan^{-1}x \text{ \& } 1 - |x|$$



$$n_3 = 1$$



6. Ans (A,B)

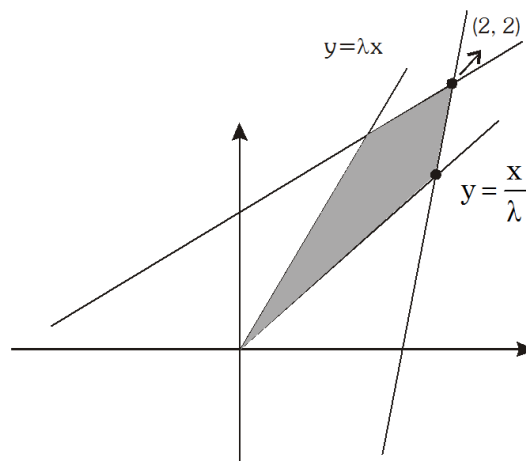
$$(x-1)^2 + (y-1)^2 + \lambda(3x-4y+1) = 0$$

$$x^2 + y^2 + (3\lambda-2)x - 2(2\lambda+1)y + (2+\lambda) = 0$$

$$-3\lambda + 2 + 2\lambda + 1 - 5 = 0 \Rightarrow -\lambda - 2 = 0 \Rightarrow \lambda = -2$$

$$x^2 + y^2 - 8x + 6y = 0 \xrightarrow[\text{Circle}]{\text{Director}} (x-4)^2 + (y+3)^2 = 50 \Rightarrow x_{\text{int}} = 2\sqrt{41}$$

7. Ans (C,D)



$$\left(\frac{y}{x}\right)^2 - \left(\frac{k}{\sqrt{3}}\right) \cdot \left(\frac{y}{x}\right) + 1 = 0$$

$$\Rightarrow \frac{y}{x} = \frac{\frac{k}{\sqrt{3}} \pm \sqrt{\frac{k^2}{3} - 4}}{2}$$

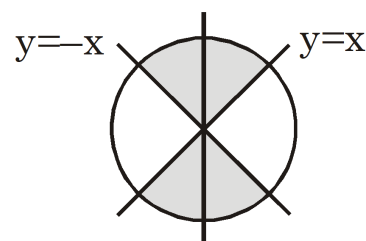
$$\frac{y}{x} = \frac{\frac{k}{\sqrt{3}} \pm \sqrt{\frac{k^2}{3} - 4}}{2} = \lambda, \quad \frac{1}{\lambda}; \quad \lambda \in [1, \sqrt{3}];$$

$$\lambda = \frac{k + \sqrt{k^2 - 12}}{2\sqrt{3}}$$

$$\min(A_k) = 0 \text{ at } k = 2\sqrt{3}$$

$$\max(A_k) = 8 - 4\sqrt{3} \text{ at } k = 4$$

8. Ans (B)



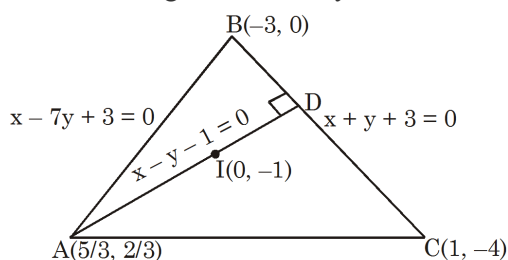
$$\text{Area}(T-S) = \frac{\pi(2)^2}{2} = 2\pi$$

9. Ans (A,C)

Solve AB and BC to get B(-3, 0)

Solve AD and AB to get A(5/3, 2/3)

Take the image of B in $x - y - 1 = 0$



(which is altitude as well as angle bisector through A) to get C(1, -4), now get AC

10. **Ans (B,C,D)**

$\because$ If $x \rightarrow 0^+ 2013^x \in (1, 2)$

$$\Rightarrow \{2013^x\} = 2013^x - 1$$

Ans as $x \rightarrow 0^- 2013^x \in (0, 1)$

$$\Rightarrow \{2013^x\} = 2013^x$$

$$\therefore \lim_{x \rightarrow 0^+} f(2013^x) = \lim_{x \rightarrow 0^+} (2013^x)^{\frac{1}{\{2013^x\}}}$$

$$= \lim_{x \rightarrow 0^+} (2013^x)^{\frac{1}{2013^x - 1}}$$

$$= e^{\lim_{x \rightarrow 0^+} \frac{1}{2013^x - 1} (2013^x - 1)} = e$$

$$\lim_{x \rightarrow 0^-} f(2013^x) = \lim_{x \rightarrow 0^-} (2013^x)^{\frac{1}{2013^x}} = 1^1 = 1$$

LHL $\neq$ RHL

11. **Ans (A,C)**

Let $g(x) = f(x) - f(x+1)$ for $x \in [0, 1]$

$$g(0) = f(0) - f(1)$$

$$g(1) = f(1) - f(2)$$

$$\therefore g(0) + g(1) = f(0) - f(2) = 0$$

$\Rightarrow$ atleast one root of $g(x) = 0$ lies in $[0, 1]$

and for other options plot graph of

$$f(x) = \sum_{r=1}^{2017} \frac{r}{x-r}$$

12. **Ans (A,B,C,D)**

$$f(x) + f(y) = f(x)f(y) + f(xy)$$

partial differentiation wrt x

$$f'(x) = f'(x)f(y) + f'(xy) \cdot (y)$$

put $x = 1$

$$-2 = -2f(y) + f'(y)y$$

$$\text{get } f(y) = 1 - y^2 \Rightarrow f(x) = 1 - x^2$$

Now, number of solution(s) of $f(x) = 2$

$$1 - x^2 = 2 \Rightarrow x^2 = -1 \Rightarrow x \in \emptyset$$

PART-3 : MATHEMATICS

SECTION-II

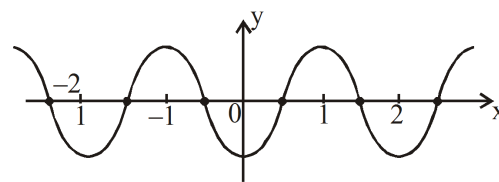
1. **Ans (1.00)**

$$f(x) = |(2^x - 1) \cdot (3^x - 3)x \cdot (x - 1) \cdot (x - 2)|$$

$f(x)$ is not differentiable at $x = 2$

2. **Ans (6.00)**

$$\lim_{x \rightarrow \infty} P(x) \rightarrow \infty \text{ and } \lim_{x \rightarrow (-\infty)} P(x) \rightarrow \infty.$$



$P(0) = -1$ and $P(x)$ is even this $P(1) = P(-1) = 1$

& $P(2) = P(-2) = -1$.

Hence, from IVT minimum no. of zeroes are 6.

3. **Ans (0.25)**

$$\tan^{-1} 3 + \tan^{-1} \frac{9}{2} + \tan^{-1} \frac{33}{4} + \dots$$

$$\frac{n\pi}{2} - \left[\tan^{-1} \frac{1}{3} + \tan^{-1} \frac{2}{9} + \tan^{-1} \frac{4}{33} + \dots \right]$$

$$\frac{n\pi}{2} \left[\tan^{-1} \left(\frac{2-1}{1+1 \cdot 2} \right) + \tan^{-1} \left(\frac{2^2-2^1}{1+2^1 \cdot 2^2} \right) \right. \\ \left. + \tan^{-1} \left(\frac{2^3-2^2}{1+2^2 \cdot 2^3} \right) + \dots \tan^{-1} \left(\frac{2^n-2^{n-1}}{1+2^{n-1} \cdot 2^n} \right) \right]$$

$$\frac{n\pi}{2} - [\tan^{-1} 2^n - \tan^{-1} 1]$$

$$\frac{(2n+1)\pi}{4} - \tan 2^n$$

4. **Ans (1.00)**

$$|f(x)| \leq |e^x - 1|$$

By taking limit $x \rightarrow -\infty$ on

Both sides

$$\lim_{x \rightarrow -\infty} |f(x)| \leq |1|$$

$$\lim_{x \rightarrow -\infty} |f(x)| \leq 1$$

$\Rightarrow$ degree of $f(x)$ cannot be ≥ 1

(Because if degree of $f(x) \geq 1$ then

$$\lim_{x \rightarrow -\infty} |f(x)| = \infty)$$

so $f(x)$ must be constant function

Now for $x = 0$

$$|f(0)| \leq |e^0 - 1|$$

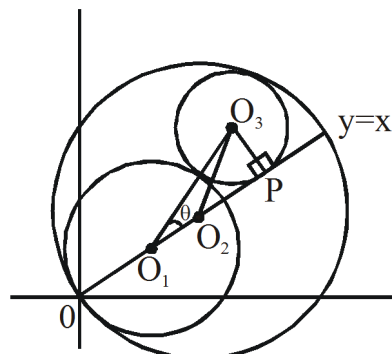
$$|f(0)| \leq 0$$

$$\therefore f(0) = 0$$

Hence $f(x) = 0 \forall x \in \mathbb{R}$

$$f'(0) = 0$$

5. Ans (1.87 or 1.88)



$$S_1 = x^2 + y^2 + 3\sqrt{2}x + 3\sqrt{2}y = 0$$

$$r = 3$$

$$S_2 : x^2 + y^2 + 5\sqrt{2}x + 5\sqrt{2}y = 0$$

$$R = 5$$

Let O_1, O_2 be centre of circle with radius 3 and 5 respectively.

Let O_3 be the centre of circle with radius x which is tangent to two given circles and common diameter

$$O_1O_2 = 5 - 3 = 2; O_1O_3 = 3 + x,$$

$$O_2O_3 = 5 - x$$

Apply cosine rule in $\triangle O_1O_2O_3$

$$\cos \theta = \frac{2^2 + (3+x)^2 - (5-x)^2}{2 \cdot 2 \cdot (3+x)} = \frac{4x-3}{3+x}$$

$$\text{Also in } \triangle O_1PO_3 \quad \sin \theta = \frac{x}{3+x}$$

$$\cos^2 \theta + \sin^2 \theta = 1$$

$$\left(\frac{4x-3}{3+x} \right)^2 + \left(\frac{x}{3+x} \right)^2 = 1$$

$$x = \frac{15}{8}$$

6. Ans (3.00)

Replace x by $\frac{x-1}{x}$, we get

$$f\left(\frac{x-1}{x}\right) + f\left(\frac{1}{1-x}\right) = \tan^{-1}\left(\frac{x-1}{x}\right) \quad \dots(2)$$

Replace x by $\frac{1}{1-x}$ in (1) we get

$$f\left(\frac{1}{1-x}\right) + f(x) = \tan^{-1}\left(\frac{1}{1-x}\right) \quad \dots(3)$$

(1) - (2) + (3), we get

$$2f(x) = \tan^{-1}x + \tan^{-1}\left(\frac{1}{1-x}\right) - \tan^{-1}\left(\frac{x-1}{x}\right) \quad \dots(4)$$

Also,

$$2f(1-x)$$

$$= \tan^{-1}(1-x) + \tan^{-1}\left(\frac{1}{x}\right) - \tan^{-1}\left(\frac{x}{x-1}\right) \quad \dots(5)$$

put $x = \frac{1}{2}$ and add (4) and (5)

$$\Rightarrow 4f\left(\frac{1}{2}\right) = \frac{3\pi}{2}$$

$$\Rightarrow N = f\left(\frac{1}{2}\right) = \frac{3\pi}{8}$$